

How to Install Hadoop with Step by Step Configuration on Linux Ubuntu

By :



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Hours Updated June 13, 2024

In this tutorial, we will take you through step by step process to install Apache Hadoop on a Linux box (Ubuntu). This is 2 part process

- [Part 1\) Download and Install Hadoop](#)
- [Part 2\) Configure Hadoop](#)

There are 2 **Prerequisites**

- You must have [Ubuntu installed](#) and running
- You must have [Java Installed](#).

Part 1) Download and Install Hadoop

Step 1) Add a Hadoop system user using below command

```
sudo addgroup hadoop_
```

```
guru99@guru99-VirtualBox:~$ sudo addgroup hadoop_  
[sudo] password for guru99:  
Adding group `hadoop_' (GID 1001) ...  
Done.
```

```
sudo adduser --ingroup hadoop_ hduser_
```

```
guru99@guru99-VirtualBox:~$ sudo adduser --ingroup hadoop_ hduser_
Adding user `hduser_' ...
Adding new user `hduser_' (1001) with group `hadoop_' ...
Creating home directory `/home/hduser_' ...
Copying files from `/etc/skel' ...
Enter new UNIX password:
Retype new UNIX password:
passwd: password updated successfully
Changing the user information for hduser_
Enter the new value, or press ENTER for the default
    Full Name []: Team
    Room Number []: 1
    Work Phone []: 1
    Home Phone []: 1
    Other []: 1
Is the information correct? [Y/n] y
guru99@guru99-VirtualBox:~$
```



Enter your password, name and other details.

NOTE: There is a possibility of below-mentioned error in this setup and installation process.

“hduser is not in the sudoers file. This incident will be reported.”

```
hduser_@ajayanand-Virtual-Ubuntu:/usr/local/newhadoop$ sudo mv hadoop-1.0.3 hadoop
[sudo] password for hduser_:
hduser_ is not in the sudoers file. This incident will be reported.
hduser_@ajayanand-Virtual-Ubuntu:/usr/local/newhadoop$ █
```

This error can be resolved by Login as a root user

```
hduser_@guru99-VirtualBox:/$ su guru99
Password:
guru99@guru99-VirtualBox:/$
```

Guru99 is
Root user

Execute the command

```
sudo adduser hduser_ sudo
```

```
guru99@guru99-VirtualBox:~$ sudo adduser hduser_ sudo
Adding user `hduser_' to group `sudo' ...
Adding user hduser_ to group sudo
Done.
```

Re-login as hduser_

```
guru99@guru99-VirtualBox:/$ su hduser_
Password:
hduser_@guru99-VirtualBox:/$
```

Step 2) Configure SSH

In order to manage nodes in a cluster, Hadoop requires SSH access

First, switch user, enter the following command

```
su - hduser_
```

```
guru99@guru99-VirtualBox:~$ su - hduser_
Password:
hduser_@guru99-VirtualBox:~$
```

This command will create a new key.

```
ssh-keygen -t rsa -P ""
```

```

guru99@guru99-VirtualBox:~$ su - hduser_
Password:
hduser_@guru99-VirtualBox:~$ ssh-keygen -t rsa -P ""
Generating public/private rsa key pair.
Enter file in which to save the key (/home/hduser_/.ssh/id_rsa):
Created directory '/home/hduser_/.ssh'.
Your identification has been saved in /home/hduser_/.ssh/id_rsa.
Your public key has been saved in /home/hduser_/.ssh/id_rsa.pub.
The key fingerprint is:
07:e2:3f:7d:7d:d1:0d:9d:12:0e:e7:27:ab:47:4a:22 hduser_@guru99-VirtualBox
The key's randomart image is:
+--[ RSA 2048 ]-----+
|           . o        |
|            = ...     |
|       . .    =.o.    |
|      . . .    =.o    |
|     .ES... o .o     |
|      ..oo +. .      |
|       o .o... .      |
|        . . . .      |
+-----+
hduser_@guru99-VirtualBox:~$

```

Press Enter

Enable SSH access to local machine using this key.

```
cat $HOME/.ssh/id_rsa.pub >> $HOME/.ssh/authorized_keys
```

```

hduser_@guru99-VirtualBox:~$ cat $HOME/.ssh/id_rsa.pub >> $HOME/.ssh/authorized_keys
hduser_@guru99-VirtualBox:~$

```

Now test SSH setup by connecting to localhost as 'hduser' user.

```
ssh localhost
```

```
hduser_@guru99-VirtualBox:/$ ssh localhost
The authenticity of host 'localhost (127.0.0.1)' can't be established.
ECDSA key fingerprint is 4a:78:f7:93:32:0a:c1:b4:24:e2:a6:78:d7:cb:20:d6.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added 'localhost' (ECDSA) to the list of known hosts.
Welcome to Ubuntu 12.04.1 LTS (GNU/Linux 3.2.0-29-generic-pae i686)

 * Documentation:  https://help.ubuntu.com/

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

hduser_@guru99-VirtualBox:~$
```

Note: Please note, if you see below error in response to ‘ssh localhost’, then there is a possibility that SSH is not available on this system-

```
hduser@guru99: ~
hduser@guru99:~$ ssh localhost
ssh: connect to host localhost port 22: Connection refused
hduser@guru99:~$
```

To resolve this –

Purge SSH using,

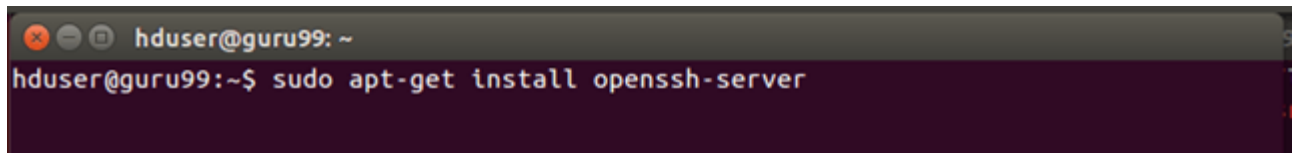
```
sudo apt-get purge openssh-server
```

It is good practice to purge before the start of installation

```
hduser@guru99:~$ sudo apt-get purge openssh-server
Reading package lists... Done
Building dependency tree
Reading state information... Done
Package 'openssh-server' is not installed, so not removed
0 upgraded, 0 newly installed, 0 to remove and 19 not upgraded.
hduser@guru99:~$
```

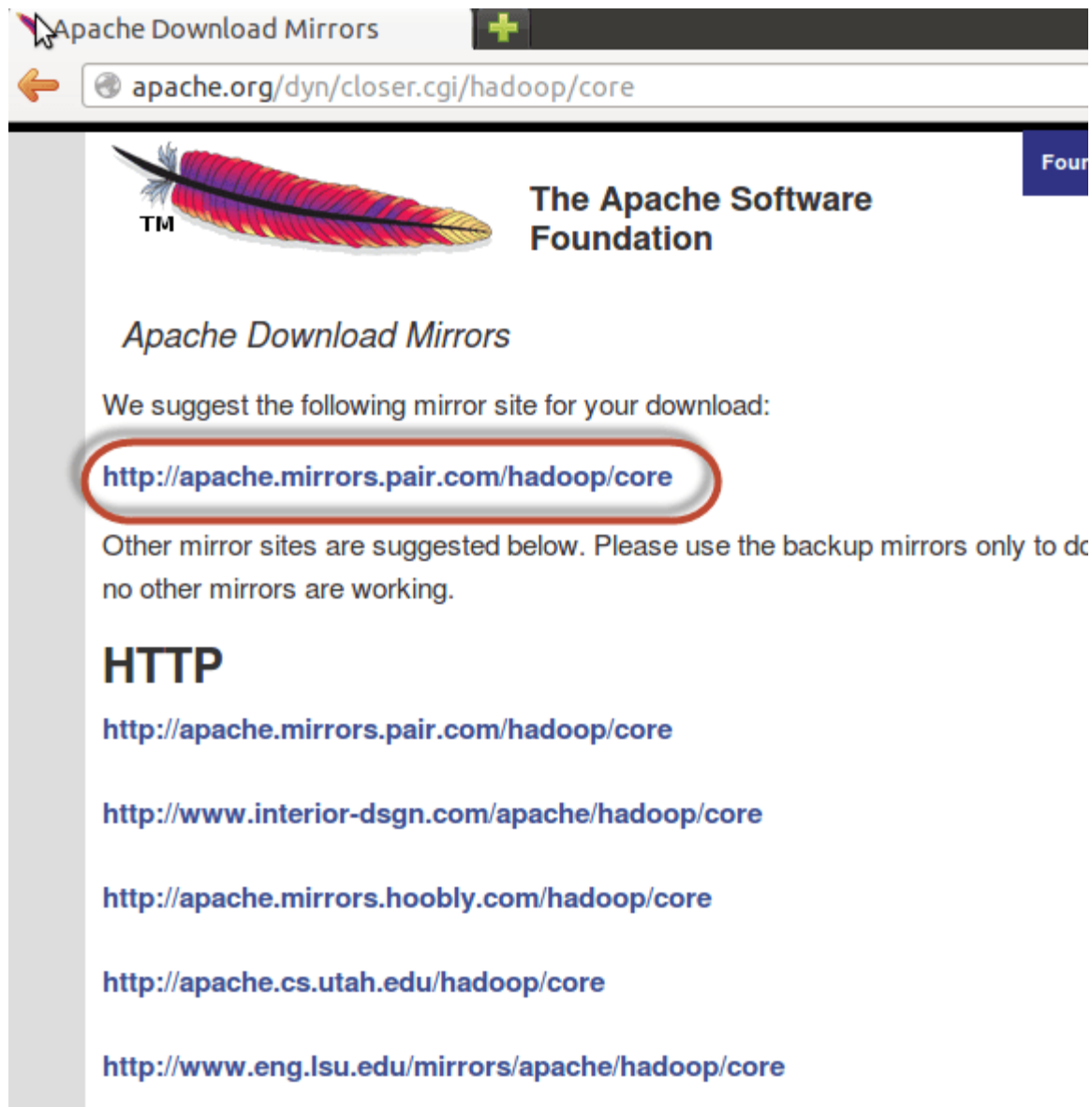
Install SSH using the command-

```
sudo apt-get install openssh-server
```

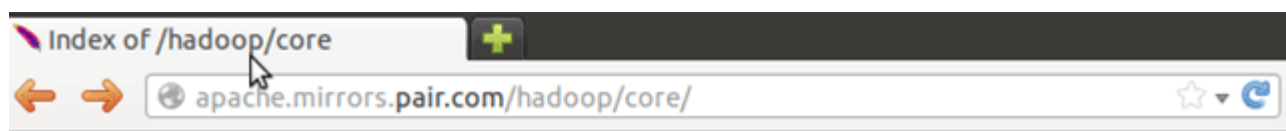


```
hduser@guru99: ~  
hduser@guru99:~$ sudo apt-get install openssh-server
```

Step 3) Next step is to [Download Hadoop](#)



Select Stable



Hadoop Releases

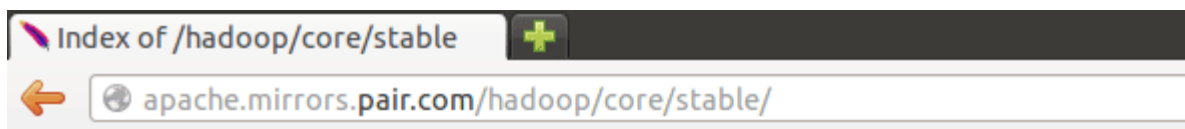
Please make sure you're downloading from [a nearby mirror site](#), not from [www.apache.org](#).

We suggest downloading the current stable release.

Older releases are available from the [archives](#).

Name	Last modified	Size	Description
Parent Directory		-	
current/	31-Mar-2014 05:19	-	
current2/	31-Mar-2014 05:19	-	
hadoop-0.23.10/	03-Dec-2013 01:07	-	
hadoop-0.23.9/	01-Jul-2013 13:16	-	
hadoop-1.2.1/	22-Jul-2013 18:49	-	
hadoop-2.0.3-alpha/	06-Feb-2013 22:53	-	

Select the tar.gz file (not the file with src)



Index of /hadoop/core/stable

Name	Last modified	Size	Description
Parent Directory		-	
hadoop-2.2.0-src.tar.gz	07-Oct-2013 02:46	19M	GZIP compressed document
hadoop-2.2.0-src.tar.gz.mds	07-Oct-2013 02:46	1.1K	GZIP compressed document
hadoop-2.2.0.tar.gz	07-Oct-2013 02:46	104M	GZIP compressed document
hadoop-2.2.0.tar.gz.mds	07-Oct-2013 02:47	958	GZIP compressed document

Once a download is complete, navigate to the directory containing the tar file

```
hduser_@guru99-VirtualBox:~$ cd /home/guru99/Downloads
```

Enter,

```
sudo tar xzf hadoop-2.2.0.tar.gz
```

```
hduser_@guru99-VirtualBox: /home/guru99/Downloads$ sudo tar -xvf hadoop-2.2.0.tar.gz
```

Now, rename hadoop-2.2.0 as hadoop

```
sudo mv hadoop-2.2.0 hadoop
```

```
hduser_@guru99-VirtualBox: /home/guru99/Downloads$ sudo mv hadoop-2.2.0 hadoop
hduser_@guru99-VirtualBox: /home/guru99/Downloads$
```

```
sudo chown -R hduser_:hadoop_ hadoop
```

```
hduser_@guru99-VirtualBox: /home/guru99/Downloads$ sudo chown -R hduser_:hadoop_ hadoop
hduser_@guru99-VirtualBox: /home/guru99/Downloads$
```

RELATED ARTICLES

- [What is Hadoop? Introduction, Architecture, Ecosystem, Components](#)
- [HDFS Tutorial: Architecture, Read & Write Operation using Java API](#)

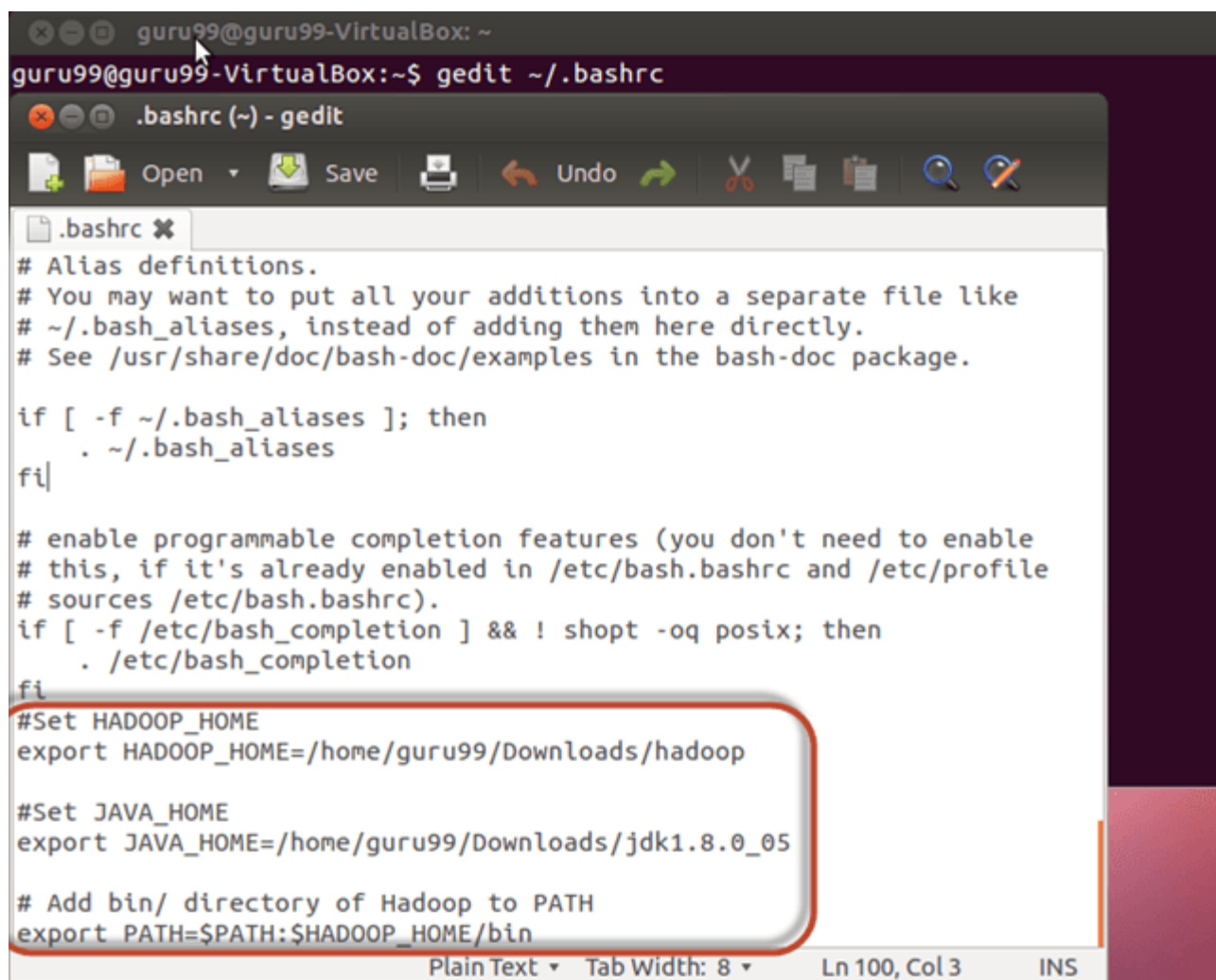
- [Hadoop Pig Tutorial: What is Apache Pig? Architecture, Example](#)
- [Talend Tutorial – What is Talend ETL Tool?](#)

Part 2) Configure Hadoop

Step 1) Modify ~/.bashrc file

Add following lines to end of file ~/.bashrc

```
#Set HADOOP_HOME
export HADOOP_HOME=<Installation Directory of Hadoop>
#Set JAVA_HOME
export JAVA_HOME=<Installation Directory of Java>
# Add bin/ directory of Hadoop to PATH
export PATH=$PATH:$HADOOP_HOME/bin
```



```
guru99@guru99-VirtualBox: ~  
guru99@guru99-VirtualBox:~$ gedit ~/.bashrc  
.bashrc (~) - gedit  
# Alias definitions.  
# You may want to put all your additions into a separate file like  
# ~/.bash_aliases, instead of adding them here directly.  
# See /usr/share/doc/bash-doc/examples in the bash-doc package.  
  
if [ -f ~/.bash_aliases ]; then  
    . ~/.bash_aliases  
fi  
  
# enable programmable completion features (you don't need to enable  
# this, if it's already enabled in /etc/bash.bashrc and /etc/profile  
# sources /etc/bash.bashrc).  
if [ -f /etc/bash_completion ] && ! shopt -oq posix; then  
    . /etc/bash_completion  
fi  
#Set HADOOP_HOME  
export HADOOP_HOME=/home/guru99/Downloads/hadoop  
  
#Set JAVA_HOME  
export JAVA_HOME=/home/guru99/Downloads/jdk1.8.0_05  
  
# Add bin/ directory of Hadoop to PATH  
export PATH=$PATH:$HADOOP_HOME/bin
```

Now, source this environment configuration using below command

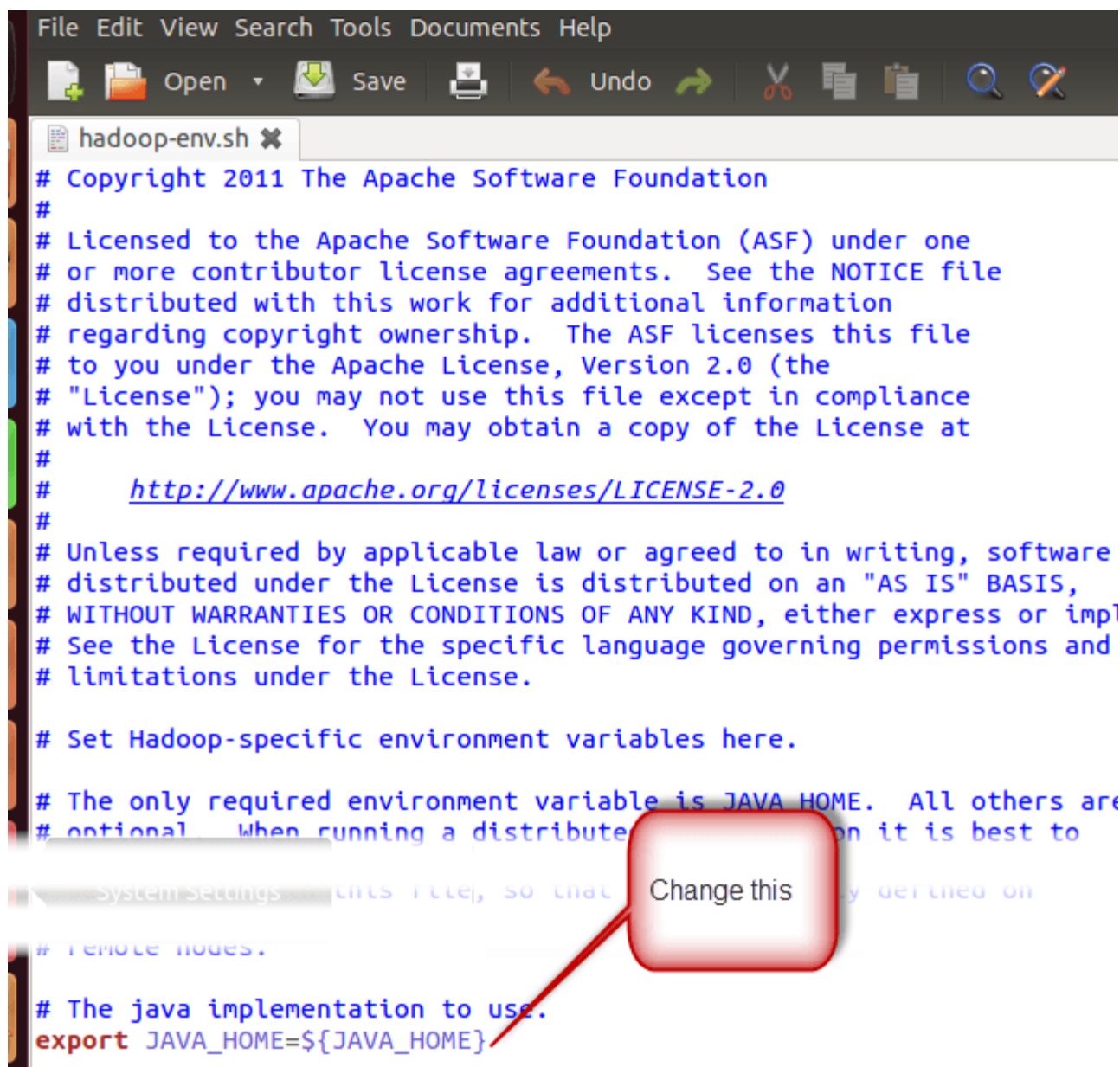
```
. ~/.bashrc
```

```
guru99@guru99-VirtualBox:~$ . ~/.bashrc  
guru99@guru99-VirtualBox:~$
```

Step 2) Configurations related to HDFS

Set **JAVA_HOME** inside file **\$HADOOP_HOME/etc/hadoop/hadoop-env.sh**

```
guru99@guru99-VirtualBox:~$ sudo gedit /home/guru99/Downloads/hadoop/etc/hadoop/hadoop-env.sh
```



```
File Edit View Search Tools Documents Help
Open Save Undo
hadoop-env.sh
# Copyright 2011 The Apache Software Foundation
#
# Licensed to the Apache Software Foundation (ASF) under one
# or more contributor license agreements. See the NOTICE file
# distributed with this work for additional information
# regarding copyright ownership. The ASF licenses this file
# to you under the Apache License, Version 2.0 (the
# "License"); you may not use this file except in compliance
# with the License. You may obtain a copy of the License at
#
# http://www.apache.org/licenses/LICENSE-2.0
#
# Unless required by applicable law or agreed to in writing, software
# distributed under the License is distributed on an "AS IS" BASIS,
# WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
# See the License for the specific language governing permissions and
# limitations under the License.
#
# Set Hadoop-specific environment variables here.
#
# The only required environment variable is JAVA_HOME. All others are
# optional. When running a distributed cluster on it is best to
#
# System Settings
# Remote nodes.
#
# The java implementation to use.
export JAVA_HOME=${JAVA_HOME}
```

With

```
# The java implementation to use.  
export JAVA_HOME=/home/guru99/Downloads/jdk1.8.0_05
```

There are two parameters in **\$HADOOP_HOME/etc/hadoop/core-site.xml** which need to be set-

1. **‘hadoop.tmp.dir’** – Used to specify a directory which will be used by Hadoop to store its data files.
2. **‘fs.default.name’** – This specifies the default file system.

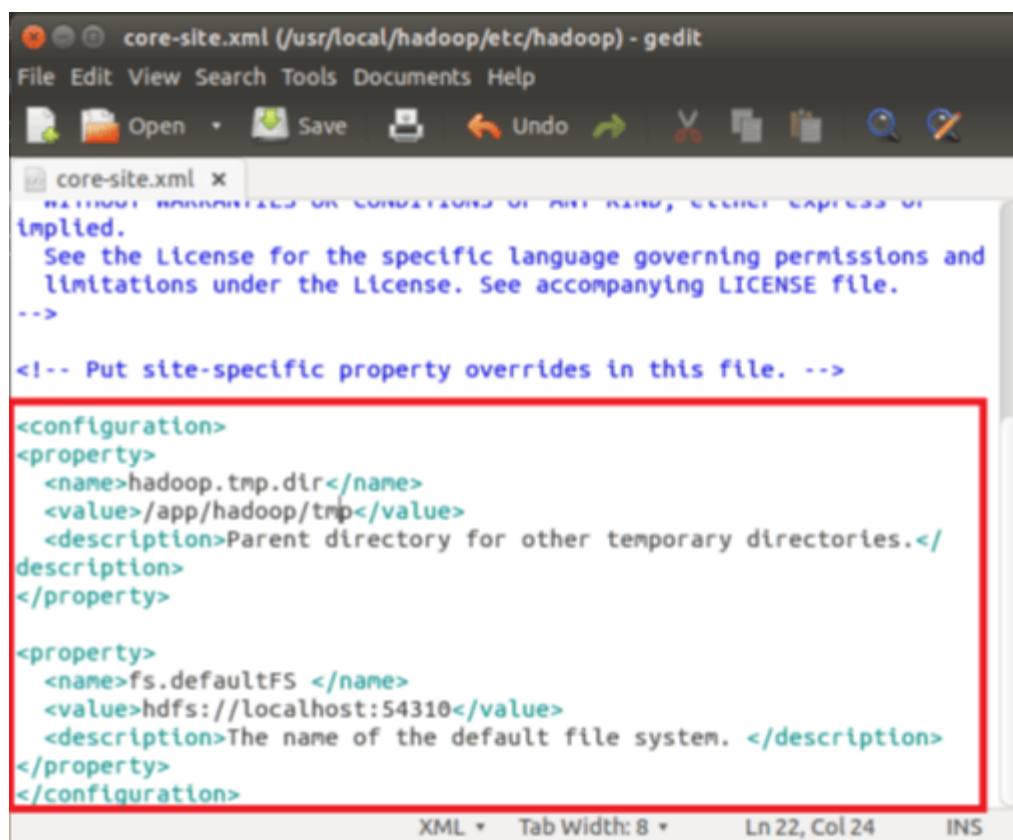
To set these parameters, open core-site.xml

```
sudo gedit $HADOOP_HOME/etc/hadoop/core-site.xml
```

```
guru99@guru99-VirtualBox:~$ sudo gedit /home/guru99/Downloads/hadoop/etc/hadoop/core-site.xml
```

Copy below line in between tags `<configuration></configuration>`

```
<property>  
<name>hadoop.tmp.dir</name>  
<value>/app/hadoop/tmp</value>  
<description>Parent directory for other temporary directories.</description>  
</property>  
<property>  
<name>fs.defaultFS </name>  
<value>hdfs://localhost:54310</value>  
<description>The name of the default file system. </description>  
</property>
```



Navigate to the directory **\$HADOOP_HOME/etc/Hadoop**

```
guru99@guru99-VirtualBox:~$ cd /home/guru99/Downloads/hadoop/etc/hadoop
guru99@guru99-VirtualBox:~/Downloads/hadoop/etc/hadoop$
```

Now, create the directory mentioned in core-site.xml

```
sudo mkdir -p <Path of Directory used in above setting>
```

```
guru99@guru99-VirtualBox:~/Downloads/hadoop/etc/hadoop$ sudo mkdir -p /app/hadoop/tmp
guru99@guru99-VirtualBox:~/Downloads/hadoop/etc/hadoop$
```

Grant permissions to the directory

```
sudo chown -R hduser_:Hadoop_ <Path of Directory created in above step>
```

```
hduser_@guru99-VirtualBox:~$ sudo chown -R hduser_:hadoop_ /app/hadoop/tmp
hduser_@guru99-VirtualBox:~$
```

```
sudo chmod 750 <Path of Directory created in above step>
```

```
hduser_@guru99-VirtualBox:~$ sudo chmod 750 /app/hadoop/tmp
hduser_@guru99-VirtualBox:~$
```

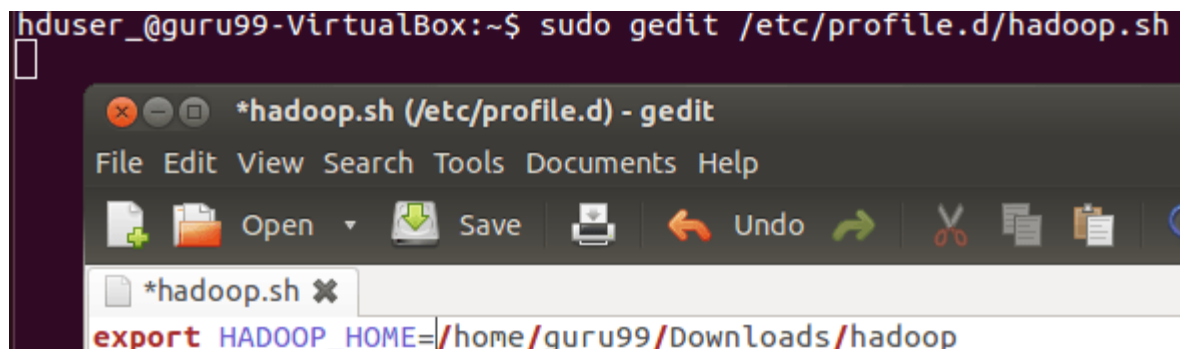
Step 3) Map Reduce Configuration

Before you begin with these configurations, lets set HADOOP_HOME path

```
sudo gedit /etc/profile.d/hadoop.sh
```

And Enter

```
export HADOOP_HOME=/home/guru99/Downloads/Hadoop
```



Next enter

```
sudo chmod +x /etc/profile.d/hadoop.sh
```

```
hduser_@guru99-VirtualBox:/$ sudo chmod +x /etc/profile.d/hadoop.sh
```

Exit the Terminal and restart again

Type echo \$HADOOP_HOME. To verify the path

```
guru99@guru99-VirtualBox:~$ echo $HADOOP_HOME  
/home/guru99/Downloads/hadoop
```

Now copy files

```
sudo cp $HADOOP_HOME/etc/hadoop/mapred-site.xml.template  
$HADOOP_HOME/etc/hadoop/mapred-site.xml
```

```
guru99@guru99-VirtualBox:~$ sudo cp $HADOOP_HOME/etc/hadoop/mapred-site.xml.temp  
late $HADOOP_HOME/etc/hadoop/mapred-site.xml
```

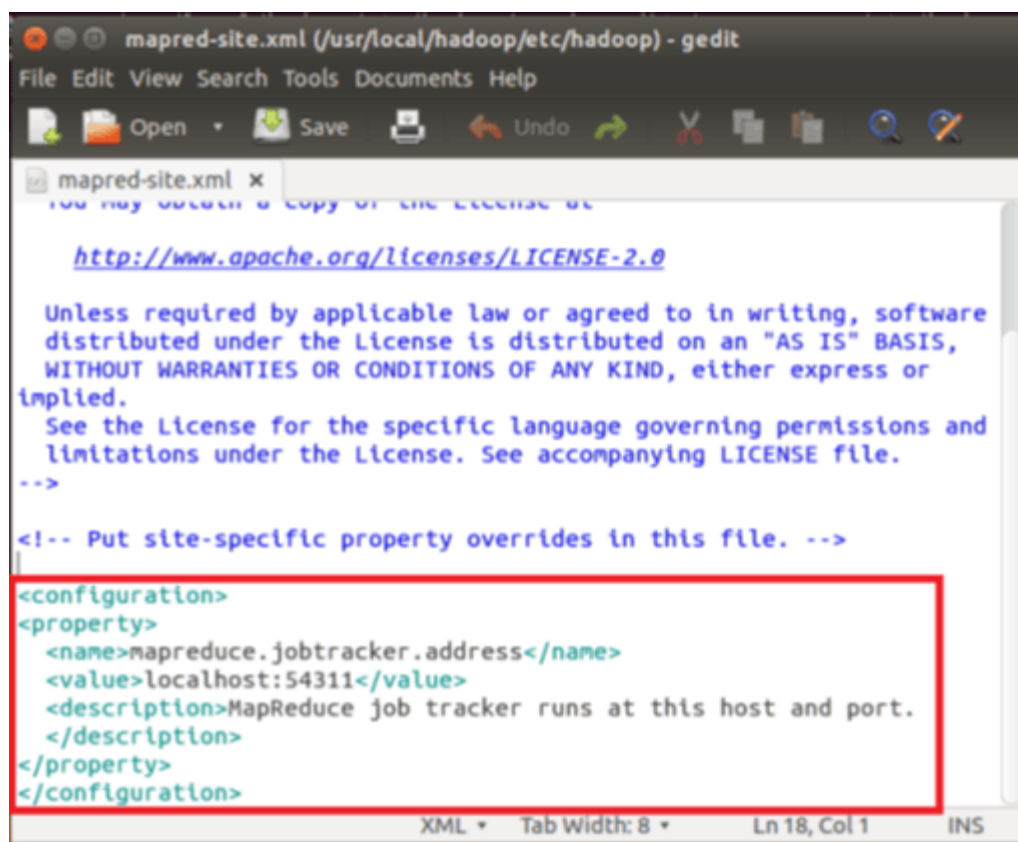
Open the **mapred-site.xml** file

```
sudo gedit $HADOOP_HOME/etc/hadoop/mapred-site.xml
```

```
guru99@guru99-VirtualBox:~$ sudo gedit $HADOOP_HOME/etc/hadoop/mapred-site.xml
```

Add below lines of setting in between tags <configuration> and </configuration>

```
<property>  
<name>mapreduce.jobtracker.address</name>  
<value>localhost:54311</value>  
<description>MapReduce job tracker runs at this host and port.  
</description>  
</property>
```

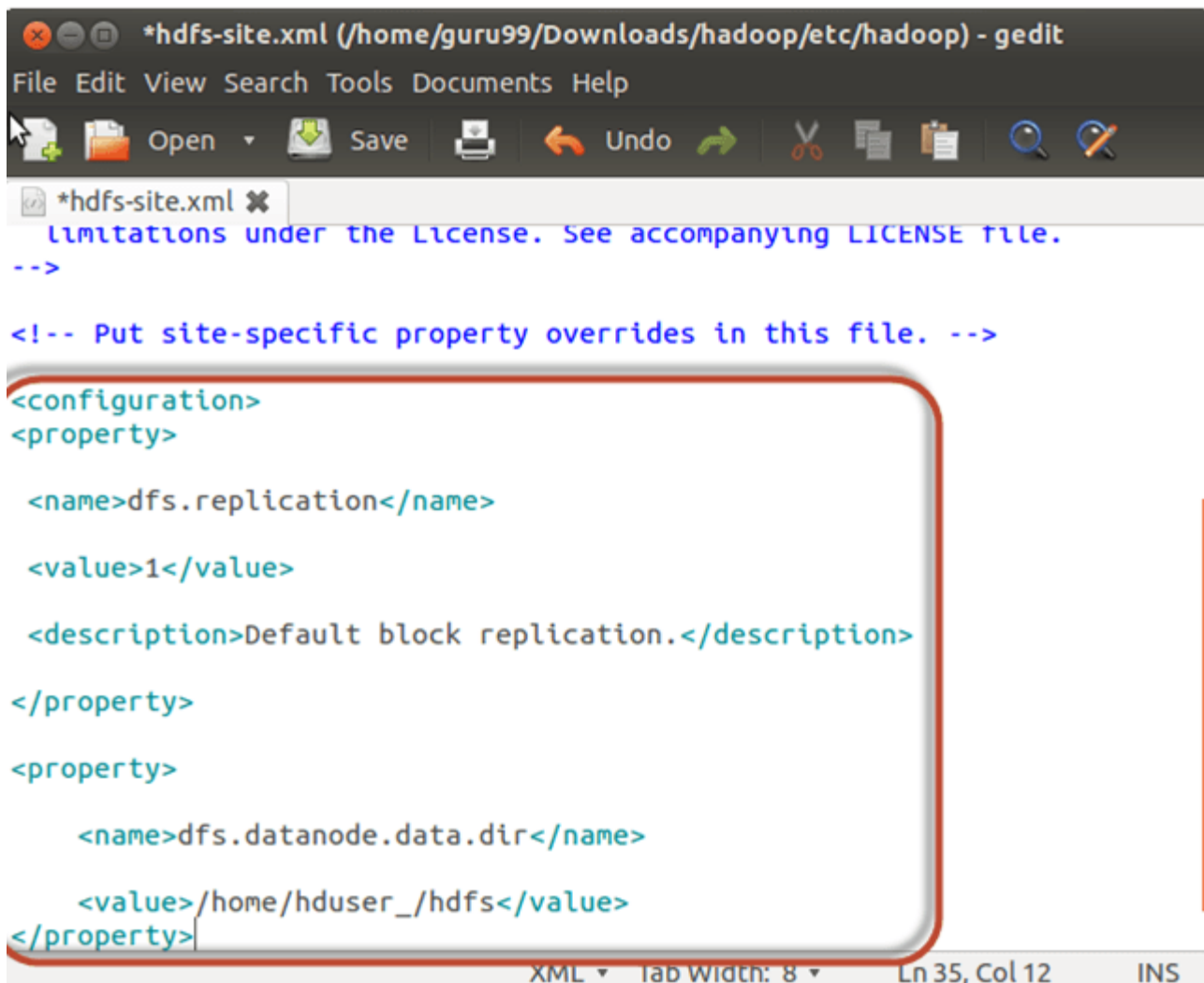
Open `$HADOOP_HOME/etc/hadoop/hdfs-site.xml` as below,

```
sudo gedit $HADOOP_HOME/etc/hadoop/hdfs-site.xml
```

```
hduser_@guru99-VirtualBox:~$ sudo gedit $HADOOP_HOME/etc/hadoop/hdfs-site.xml
```

Add below lines of setting between tags `<configuration>` and `</configuration>`

```
<property>
<name>dfs.replication</name>
<value>1</value>
<description>Default block replication.</description>
</property>
<property>
<name>dfs.datanode.data.dir</name>
<value>/home/hduser_/hdfs</value>
</property>
```



```
*hdfs-site.xml (/home/guru99/Downloads/hadoop/etc/hadoop) - gedit
File Edit View Search Tools Documents Help
Open Save Undo
*hdfs-site.xml x
limitations under the License. See accompanying LICENSE file.
-->

<!-- Put site-specific property overrides in this file. -->

<configuration>
<property>

  <name>dfs.replication</name>

  <value>1</value>

  <description>Default block replication.</description>
</property>

<property>

  <name>dfs.datanode.data.dir</name>

  <value>/home/hduser_/hdfs</value>
</property>
```

Create a directory specified in above setting-

```
sudo mkdir -p <Path of Directory used in above setting>
```

```
sudo mkdir -p /home/hduser_/hdfs
```

```
hduser_@guru99-VirtualBox:~$ sudo mkdir -p /home/hduser_/hdfs
```

```
sudo chown -R hduser_:hadoop_ <Path of Directory created in above step>
```

```
sudo chown -R hduser_:hadoop_ /home/hduser_/hdfs
```

```
hduser_@guru99-VirtualBox:~$ sudo chown -R hduser_:hadoop_ /home/hduser_/hdfs
```

```
sudo chmod 750 <Path of Directory created in above step>
```

```
sudo chmod 750 /home/hduser_/hdfs
```

```
hduser_@guru99-VirtualBox:~$ sudo chmod 750 /home/hduser_/hdfs
```

Step 4) Before we start Hadoop for the first time, format HDFS using below command

```
$HADOOP_HOME/bin/hdfs namenode -format
```

```
hduser_@guru99-VirtualBox:~$ $HADOOP_HOME/bin/hdfs namenode -format
14/05/05 13:01:58 INFO namenode.NameNode: STARTUP_MSG:
/*****
STARTUP_MSG: Starting NameNode
STARTUP_MSG:   host = guru99-VirtualBox/127.0.1.1
STARTUP_MSG:   args = [-format]
STARTUP_MSG:   version = 2.2.0
STARTUP_MSG:   classpath = /home/guru99/Downloads/hadoop/etc/hadoop:/home/guru99
/Downloads/hadoop/share/hadoop/common/lib/activation-1.1.jar:/home/guru99/Downlo
ads/hadoop/share/hadoop/common/lib/netty-3.6.2.Final.jar:/home/guru99/Downloads/
hadoop/share/hadoop/common/lib/protobuf-java-2.5.0.jar:/home/guru99/Downloads/ha
doop/share/hadoop/common/lib/xmlenc-0.52.jar:/home/guru99/Downloads/hadoop/share
/hadoop/common/lib/jsp-api-2.1.jar:/home/guru99/Downloads/hadoop/share/hadoop/co
mmon/lib/commons-collections-3.2.1.jar:/home/guru99/Downloads/hadoop/share/hadoo
p/common/lib/avro-1.7.4.jar:/home/guru99/Downloads/hadoop/share/hadoop/common/li
b/jackson-core-asl-1.8.8.jar:/home/guru99/Downloads/hadoop/share/hadoop/common/l
ib/commons-io-2.1.jar:/home/guru99/Downloads/hadoop/share/hadoop/common/lib/jers
ey-core-1.9.jar:/home/guru99/Downloads/hadoop/share/hadoop/common/lib/commons-co
dec-1.4.jar:/home/guru99/Downloads/hadoop/share/hadoop/common/lib/mockito-all-1.
8.5.jar:/home/guru99/Downloads/hadoop/share/hadoop/common/lib/commons-cli-1.2.ja
r:/home/guru99/Downloads/hadoop/share/hadoop/common/lib/jets3t-0.6.1.jar:/home/g
uru99/Downloads/hadoop/share/hadoop/common/lib/guava-11.0.2.jar:/home/guru99/Dow
nloads/hadoop/share/hadoop/common/lib/commons-net-3.1.jar:/home/guru99/Downloads
/hadoop/share/hadoop/common/lib/commons-httpclient-3.1.jar:/home/guru99/Download
```

Step 5) Start Hadoop single node cluster using below command

```
$HADOOP_HOME/sbin/start-dfs.sh
```

An output of above command

```
@guru99-VirtualBox: ~
hduser_@guru99-VirtualBox:~$ $HADOOP_HOME/sbin/start-dfs.sh
Starting namenodes on [localhost]
localhost: starting namenode, logging to /home/guru99/Downloads/hadoop/logs/hadoop-hduser_-namenode-guru99-VirtualBox.out
localhost: starting datanode, logging to /home/guru99/Downloads/hadoop/logs/hadoop-hduser_-datanode-guru99-VirtualBox.out
Starting secondary namenodes [0.0.0.0]
The authenticity of host '0.0.0.0 (0.0.0.0)' can't be established.
ECDSA key fingerprint is 4a:78:f7:93:32:0a:c1:b4:24:e2:a6:78:d7:cb:20:06.
Are you sure you want to continue connecting (yes/no)? yes
0.0.0.0: Warning: Permanently added '0.0.0.0' (ECDSA) to the list of known hosts.
0.0.0.0: starting secondarynamenode, logging to /home/guru99/Downloads/hadoop/logs/hadoop-hduser_-secondarynamenode-guru99-VirtualBox.out
hduser_@guru99-VirtualBox:~$
```

`$HADOOP_HOME/sbin/start-yarn.sh`

```
hduser_@guru99-VirtualBox:~$ $HADOOP_HOME/sbin/start-yarn.sh
starting yarn daemons
starting resourcemanager, logging to /home/guru99/Downloads/hadoop/logs/yarn-hduser_-resourcemanager-guru99-VirtualBox.out
localhost: starting nodemanager, logging to /home/guru99/Downloads/hadoop/logs/yarn-hduser_-nodemanager-guru99-VirtualBox.out
hduser_@guru99-VirtualBox:~$
```

Using ‘**jps**’ tool/command, verify whether all the Hadoop related processes are running or not.

```
hduser_@guru99-VirtualBox:~$ jps
3732 SecondaryNameNode
4326 Jps
3865 ResourceManager
3466 DataNode
4061 NodeManager
3279 NameNode
hduser_@guru99-VirtualBox:~$
```

If Hadoop has started successfully then an output of jps should show NameNode, NodeManager, ResourceManager, SecondaryNameNode, DataNode.

Step 6) Stopping Hadoop

`$HADOOP_HOME/sbin/stop-dfs.sh`

```
hduser_@guru99-VirtualBox:~$ $HADOOP_HOME/sbin/stop-dfs.sh
Stopping namenodes on [localhost]
localhost: stopping namenode
localhost: stopping datanode
Stopping secondary namenodes [0.0.0.0]
0.0.0.0: stopping secondarynamenode
hduser_@guru99-VirtualBox:~$
```

`$HADOOP_HOME/sbin/stop-yarn.sh`

```
hduser_@guru99-VirtualBox:~$ $HADOOP_HOME/sbin/stop-yarn.sh
stopping yarn daemons
stopping resourcemanager
localhost: stopping nodemanager
no proxyserver to stop
hduser_@guru99-VirtualBox:~$
```

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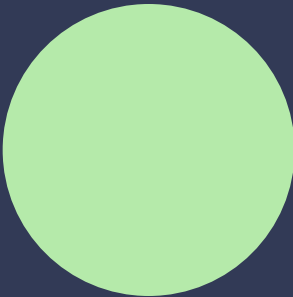
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